Memory Circuits & Systems

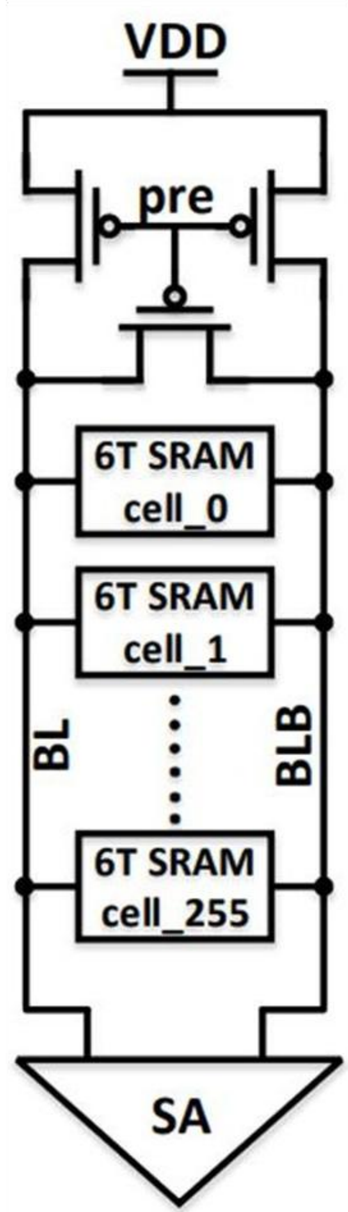
Exercise 3
Sensing Amplifier

Professor Po-Tsang Huang

Institute of Electronics
National Yang Ming Chiao Tung University

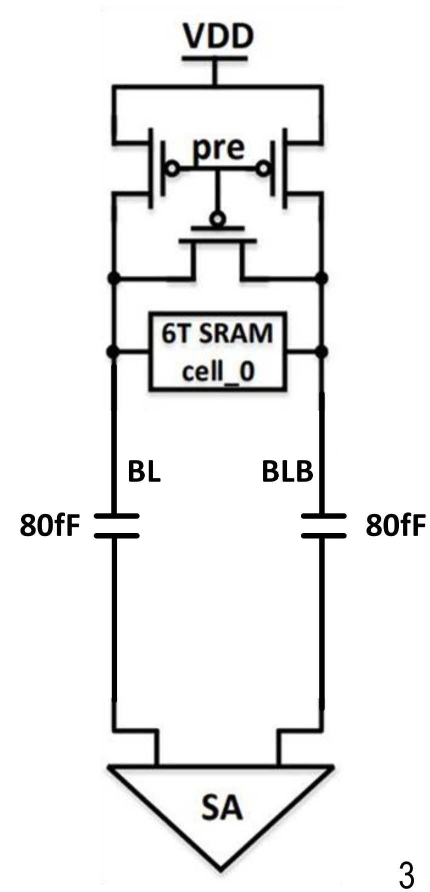
Read mode of SRAM

- Step 1:
  - BL and BLB precharged by precharger
- Step 2:
  - Activate WL of the selected row
- Step 3:
  - Depend on the data of the selected cell, BL or BLB discharged by the pull down NMOS
- Step 4:
  - Enable the sense amplifier
- Step 5:
  - Sense read data



1 . Design & Analysis of Sensing Amplifier

- During SRAM read operation, sensing amplifiers are used to detect small voltage differences between BL and BLB. In this problem, you are required to **implement the SRAM read operation** using a 7nm FinFET model. **Two types of sensing amplifiers(SA) must be implemented.**
  - Input: PRE, WL, SA_EN (**These input signals are defined by TA and cannot be modified**)
  - Output: sense
  - **Setup**
    - **1 SRAM cell with BL/BLB capacitance= 80 fF** *(Assumed wire loading of 1 column with 512 SRAM cells)*
    - Two types of sensing amplifiers must be designed:
      - Current-mode sensing amplifier
      - Voltage-mode sensing amplifier
  - **Compare and analyze the performance of the 2 SA in terms of read power and sensing timing**



2 . Robustness under Bitline Voltage Offset

- During SRAM read operation, small voltage differences between BL and BLB are used for sensing. However, mismatch and non-ideal precharge may introduce a bitline voltage offset. In this problem, you are required to design and implement the SRAM read operation using a 7nm FinFET model, and **ensure correct read operation under BL/BLB voltage offset. Monte Carlo simulation must be used** to evaluate the robustness under random bitline voltage offset.
  - Input: PRE, WL, SA_EN (**These input signals are defined by TA and cannot be modified**)
  - Output: sense
  - **Setup:**
    - **BL/BLB voltage offset :**
      - **BL/BLB = VDD $\pm$ ΔV**
      - **ΔV follows Gaussian distribution (σ = 20 mV)**
    - 1 SRAM cell with BL/BLB capacitance= 80 fF (wire loading)
  - **Monte Carlo Simulation:**
    - Number of runs: **1000**
    - **Read accuracy should achieve 100%**

3 . Robustness under V_{th} Variation

- The sensing amplifier relies on a cross-coupled latch to amplify small voltage differences between BL and BLB. However, threshold voltage (V_{th}) variation of transistors may introduce mismatch in the latch, potentially leading to incorrect sensing results. In this problem, you are required to design and implement the SRAM read operation using a 7nm FinFET model, and **ensure correct read operation under Vth variation in the sensing amplifier**. **Monte Carlo simulation must be used** to evaluate the sensing robustness under transistor V_{th} variation
  - Input: PRE, WL, SA_EN (**These input signals are defined by TA and cannot be modified**)
  - Output: sense
  - **Setup:**
    - **V_{th} variation** (σ_v_{th} follows Gaussian distribution) **is applied to the cross-coupled latch (4 transistors) in the sensing amplifier**
      - Case1 : **σ_v_{th} = 0.01 V**
      - Case2 : **increase the σ_v_{th} to 0.015 V**
    - 1 SRAM cell with BL/BLB capacitance= 80 fF (wire loading)
  - **Monte Carlo Simulation**
    - Number of runs: **1000**
    - Case 1: **Read accuracy must achieve 100%**
    - Case 2: **Analyze the degradation in read accuracy and identify the root cause of sensing failure. Propose possible approaches to improve robustness**

Example: Get Read accuracy from .mt file

- For each Monte Carlo simulation, the number of read “0” success/1000 is the read accuracy of the simulation.
- The read accuracy shown below is 99.8%

> Users > User > AppData > Roaming > MobaXterm > slash > RemoteFiles > 329350_2_43 > vmsa.mt0

1.0000000 Aa ab ? of 998

```
$DATA1 SOURCE='PrimeSim HSPICE' VERSION='T-2022.06 linux64' PARAM_COUNT=1
$OPTION MEASFORM=1
$index=1 is a nominal sample in Monte Carlo.
.TITLE '.title voltage mode sensing amplifier'
index tp avg_read_pwr v_final read0_success temper alter#
1  2.477339e-11  3.412663e-06  1.999422e-04    1.0000000   25.0000000 1
2  2.479158e-11  3.415395e-06  1.955878e-04    1.0000000   25.0000000 1
3  2.484849e-11  3.413874e-06  1.921817e-04    1.0000000   25.0000000 1
4  2.435345e-11  3.397911e-06  2.220704e-04    1.0000000   25.0000000 1
5  2.507516e-11  3.416729e-06  2.082104e-04    1.0000000   25.0000000 1
6  2.426686e-11  3.400179e-06  2.279978e-04    1.0000000   25.0000000 1
7  2.450411e-11  3.401060e-06  2.180111e-04    1.0000000   25.0000000 1
8  2.603479e-11  3.416151e-06  3.753424e-04    1.0000000   25.0000000 1
9  2.564849e-11  3.393279e-06  3.236034e-04    1.0000000   25.0000000 1
10  2.440431e-11  3.398932e-06  2.154594e-04    1.0000000   25.0000000 1
11  2.506007e-11  3.412451e-06  2.304609e-04    1.0000000   25.0000000 1
12  2.456089e-11  3.403393e-06  2.066741e-04    1.0000000   25.0000000 1
13  2.474282e-11  3.411282e-06  2.000083e-04    1.0000000   25.0000000 1
14  2.780516e-11  3.397292e-06  5.217873e-04    1.0000000   25.0000000 1
15  2.579686e-11  3.401234e-06  3.303800e-04    1.0000000   25.0000000 1
16  2.656064e-11  3.429944e-06  4.713984e-04    1.0000000   25.0000000 1
17  2.420092e-11  3.389960e-06  2.308850e-04    1.0000000   25.0000000 1
18  2.480175e-11  3.413516e-06  1.975672e-04    1.0000000   25.0000000 1
19  2.524809e-11  3.411797e-06  2.493004e-04    1.0000000   25.0000000 1
20  2.485160e-11  3.416823e-06  1.834523e-04    1.0000000   25.0000000 1
21  2.519402e-11  3.408464e-06  2.537402e-04    1.0000000   25.0000000 1
22  2.445848e-11  3.386913e-06  2.228291e-04    1.0000000   25.0000000 1
23  2.460053e-11  3.401881e-06  2.177356e-04    1.0000000   25.0000000 1
24  2.524724e-11  3.412091e-06  2.438352e-04    1.0000000   25.0000000 1
25  2.633048e-11  3.452108e-06  4.355922e-04    1.0000000   25.0000000 1
26  2.515742e-11  3.410745e-06  2.438389e-04    1.0000000   25.0000000 1
27  2.712224e-11  3.412650e-06  5.087618e-04    1.0000000   25.0000000 1
28  2.457693e-11  3.401914e-06  2.252926e-04    1.0000000   25.0000000 1
```

Waveform format

- Screenshot of waveform must include:
  - V(BL), V(BLB)
  - V(PRE), V(WL), V(SA_EN)
  - V(sense), V(sense_b)

Submission on E3 platform

- Please compress your **report** & **source codes in a single compressed file (.zip/.rar)** and upload this single file on E3 platform
  - **Report: Simulation results, comparison, analysis and the waveforms .**
  - Naming rules of the compressed file
  - Upload file: **Ex3_#ID.zip (including report & 3 source codes)**
- Due date: 5/8 P.M. 23:55